

Project Progress Evaluation Proforma

Project-I (PROJ-CS-302A)

Date: April 1, 2026 Project Title: AI Study Buddy Project Group No.: [Group Number]

Supervisor's Name: [Supervisor Name]

Name of Project Lab Teacher: [Teacher Name]

Team Members:

Sr. No	Roll No.	Name	Project Lab Attendance (%)
1.	[Roll No 1]	[Student 1 Name]	[Attendance 1]
2.	[Roll No 2]	[Student 2 Name]	[Attendance 2]
3.	[Roll No 3]	[Student 3 Name]	[Attendance 3]
4.	[Roll No 4]	[Student 4 Name]	[Attendance 4]

Functionalities of the System Completed:

- User Authentication & Dashboard:**
 - Secure user registration and login implemented using Clerk.
 - Comprehensive "Bento Grid" style Pro Dashboard created to visualize user progress, streaks, and study stats.
- Advanced AI Integration (Core Features):**
 - AI Study Chat (RAG):** Context-aware chatbot utilizing Groq API developed to answer queries based on user notes.
 - Auto-Summarization:** Feature integrated to instantly generate concise summaries of long study materials.
 - Smart Quiz Generator:** Logic built for automatically generating revision questions.
- Smart Scheduling & Analytics:**
 - Visual contribution graph (calendar) integrated to track daily study consistency.
 - Confidence-Based Revision system built using Spaced Repetition algorithms.
- Focus & Productivity Tools:**
 - Zen Mode & Focus Mode:** Distraction-free interfaces developed with ambient backgrounds and an immersive clock.
 - Study Logger:** Wizard-style logging flow created to track active study sessions.
- Backend & Performance:**
 - REST API developed using Node.js and Express.js.
 - MongoDB integration established for persistent data storage.
 - Hybrid Redis caching mechanism (Upstash/Local) implemented to optimize database load and API speeds.

Sources used with References:

- Frontend Development:** React.js (via Vite), Tailwind CSS for styling, and Framer Motion for UI animations.
- Backend Architecture:** Node.js, Express.js.
- Database & Caching:** MongoDB (Mongoose), Redis (Upstash / ioredis).
- AI Models & SDKs:** OpenAI SDK configured for Groq API integration (LLMs).

5. **Authentication & Security:** Clerk Auth System (`@clerk/clerk-react` and `@clerk/clerk-sdk-node`).

6. **Data Visualization:** Recharts library for rendering charts and graphs.

Additional Comments: The project is currently functioning properly with the core stack (MERN + AI). The local testing across both the frontend (Vitest) and backend (Jest) is successful. The core functionalities are mapped out and UI components have been fully integrated.

Signature of the Supervisor

Signature of the Project Lab Teacher